



MODULE NAME:	MODULE CODE:
INFORMATION SYSTEMS 2A	INSY6211/w
SYSTEM ANALYSIS AND DESIGN	SAND6221/p/w

ASSESSMENT TYPE: ASSIGNMENT 2 (PAPER ONLY)

TOTAL MARK ALLOCATION: 100 MARKS

TOTAL HOURS: 10 HOURS

By submitting this assignment, you acknowledge that you have read and understood all the rules as per the terms in the registration contract, in particular the assignment and assessment rules in The IIE Assessment Strategy and Policy (IIE009), the intellectual integrity and plagiarism rules in the Intellectual Integrity and Property Rights Policy (IIE023), as well as any rules and regulations published in the student portal.

INSTRUCTIONS:

1. ***No material may be copied from sources, even if referenced correctly, unless it is a direct quote indicated with quotation marks. No more than 10% of the assignment may consist of direct quotes.***
2. ***Your assignment must be submitted through Turnitin.***
3. ***Save a copy of your assignment before submitting it.***
4. ***Assignments must be typed unless otherwise specified.***
5. ***All work must be adequately and correctly referenced.***
6. ***This is an individual assignment.***

Referencing Rubric

Providing evidence based on valid and referenced academic sources is a fundamental educational principle and the cornerstone of high-quality academic work. Part of achieving this quality is referencing in a way that is consistent and congruent with the requirements of the referencing style being used.

Therefore, inconsistent and/or incongruent referencing will result in a penalty of **a maximum of ten percent being deducted from the overall percentage** awarded to your assessment submission.

Please note that **evidence of plagiarism** in the form of copied or unreferenced work, absent reference lists, or exceptionally poor referencing **may result in action being taken in accordance with The IIE's Intellectual Integrity and Property Rights Policy (IIE023)**. Similarly, **evidence of excessive AI usage may result in action being taken in accordance with The IIE's Student Conduct, Discipline and Safety Policy (IIE015)**.

Markers are required to provide feedback to students by **circling/underlining the information in the table below that best describes the student's work and by adding constructive commentary where appropriate**. The examples provided are not exhaustive but illustrate the errors.

Deductions

- Where the student's work contains **five or more errors** aligned to the **minor errors column** below, **deduct 5% from the overall percentage**.
- Where the student's work contains **five or more errors** aligned to the **major errors column** below, **deduct 10% from the overall percentage**.
- Where **both minor and major errors** (e.g. two minor and three major, etc.) are present, **deduct 10% only** (and not 5% or 15%) from the overall percentage.

Required: Consistent and congruent referencing	Minor errors Deduct 5% from overall percentage. Example: if the response receives 70%, deduct 5%. The final mark is 65%.	Major errors Deduct 10% from the overall percentage. Example: if the response receives 70%, deduct 10%. The final mark is 60%.
Consistency - The correct referencing style for the discipline – i.e., either Harvard, OR APA (for Psychology), OR Law, OR IEEE (for ICT/Engineering) – has been used consistently for all in-text references and in the bibliography/reference list. Concepts and ideas that are quoted and/or paraphrased are referenced consistently throughout. Position of the in-text reference: an in-text reference is positioned consistently where appropriate for every quote and paraphrase.	Minor inconsistencies: - The referencing style used is generally consistent with what is required, but there are one or two changes/errors in the format of in-text referencing and/or in the bibliography/reference list. - For example, page numbers for direct quotes in-text have been provided for one source, but not in another. Or, two book chapters in the bibliography/reference list have been referenced in two different formats. Or, the publication year has been placed after the author name in one bibliography/reference list entry, and after the source title in another, etc. - Concepts and ideas in quotes and/or paraphrases are typically referenced, but a full in-text reference is missing or incomplete from one or two small sections of the work. - Position of the references: in-text references are only given at the beginning and/or end of every paragraph.	Major inconsistencies: - Poor and wholly inconsistent referencing style used in-text and/or in the bibliography/reference list. - Multiple referencing styles for the same source types have been used. - For example, the format for direct quotes in-text and/or book chapters in the bibliography/reference list and/or year of publication in the bibliography/reference list is different across multiple instances. - Concepts and ideas in quotes and/or paraphrases are haphazardly referenced in-text. - Position of the references: in-text references are only given at the beginning or end of large sections of work.
Feedback on referencing consistency:		
Congruency Each source reflected within in-text references is included accurately in the bibliography/reference list. - All bibliography/reference list entries are in the required order for the referencing style used (e.g. alphabetical, alphabetical under subheadings, numerical). - All direct quotes and paraphrases have been integrated appropriately into the text using introductory phrases, accurate grammar, etc.	Minor incongruences: - There is largely a match between the sources presented in-text and those in the bibliography/reference list, but one or two sources that appear in-text do not appear in the bibliography/reference list, or vice versa. Or key source information is missing from one or two in-text references or bibliography/reference list entries only (e.g. publication year, city of publication, URL date accessed, etc.). - There is a clear and largely accurate ordering of sources in the bibliography/reference list as required by the referencing style used, but with one or two references out of order. - An attempt has been made for source integration into the text using appropriate introductory phrases and grammar, but one or two quotes or paraphrases do not flow as clearly or logically within the sentence structure as they could.	Major incongruences: - No relationship/several incongruities between the in-text referencing and the bibliography/reference list. - For example, multiple sources are included in-text, but not in the bibliography, and/or vice versa. Key source information is missing from multiple in-text references and/or reference list entries. A URL link, rather than the actual reference, is provided in the bibliography. Sources are repeated in the reference list, etc. - Most sources are listed in a haphazard order throughout the bibliography/reference list. - Few to no appropriate introductory phrases or rules of grammar have been applied, and many direct quotes and/or paraphrases feel disconnected from the flow of the text.
Feedback on referencing congruency:		
Overall feedback on referencing, with suggested improvements:		

At the end of this specific assignment, students should be able to:

- LU1 – LO4: Identify key documents used when planning a project.
- LU2 – LO2: Explain the difference between functional and non-functional requirements.
- LU2 - LO9: Interpret and create an entity relationship diagram.
- LU3 - LO5: Explain the concepts of user interface, User Experience and Usability.
- LU3 - LO7: Discuss the impact of visibility and affordance on usability.
- LU4 - LO3: Describe the key features of Agile development.
- LU4 - LO5: Describe the Agile approach to the project management knowledge areas.

Question 1**(Marks: 10)**

Read the following scenario. Then, answer all questions in this assignment based on the information contained in the scenario. If you make any assumptions when answering a question, please state them clearly.

Scenario: Airport Customer Relations Management System

A local airport has recently upgraded its infrastructure to be able to accommodate international flights. Airport management is expecting an influx of tourists. To ensure the all-round success of this new endeavour, a heavy emphasis is being placed on keeping visitors to the airport as satisfied with the airport and its amenities as possible. To aid in maintaining good customer relations, the airport has decided to invest in the development of a new Customer Relations Management System, which would be accessible online via kiosks strategically placed around the airport as well as a downloadable mobile application.

The new system is envisioned to offer customers the following:

1. **Information: updates on flights**, airport amenities and services available at the airport.
2. **Navigation capability: GPS** navigation to various amenities and boarding gates.
3. **Instant connection to medical** and ambulatory services for any medical emergency.

You have been awarded the contract for the development of the Airport Customer Relations Management System. At the contract signing, you heard a rumour that one of the executives is known for being indecisive and therefore changes his mind frequently. He is also responsible for ultimately signing the new system into operation, so as far as possible, you will need to accommodate his changes. In the same breath, you have been told that should you deliver this system earlier than anticipated, you will receive a sizable bonus. The one thing, though, that has been constant in all of your communications with the various stakeholders is that due to the size of the investment in the project, there is no room for failure, and the system has to be at the top of its class.

Q.1.1	Create a System Vision Document for the Airport Customer Relationship Management System.	(10)

Question 2 (Marks: 10)

Q.2.1 Identify **three functional requirements** for the Airport Customer Relationship Management System. (6)

Q.2.2 Propose two **non-functional requirements** for the Airport Customer Relationship Management System. (4)

Question 3 (Marks: 45)

Assume that the following business rules are in place:

- One passenger can request the location of one and only one amenity at a time.
- It is possible for a passenger to have multiple flights departing from the same airport and, as such, can view multiple flight bookings.
- One passenger can experience multiple medical emergencies and, as such, can request medical assistance multiple times.

Q.3.1 Create an Entity Relationship Diagram that will represent the business rules presented at the beginning of the question. Include at least three attributes for every entity. Also, indicate the primary keys for each entity. (23)

Q.3.2 Select one functionality offered by the Airport Customer Relationship Management System. Then, propose interfaces for the use of that functionality. Any software can be used to create the interfaces. Include screenshots of the interfaces in your design. (22)

Once you have designed the interfaces, discuss how you addressed the following concepts in your designs:

- Usability
- Visibility
- Affordance.

Question 4**(Marks: 35)**

The questions below relate to the following case study:

System: Sales Accounts Management System

Reciprocal Sales Corporation is a global sales company that works with leading Fortune 500 corporations. The company offers sales consultants who assist Fortune 500 corporations with their product sales, marketing and advertising. Due to the rapid growth, a Software Development department was established to improve and automate all the manual processes and workflows. The department is currently conducting a study in order to develop a new Sales Accounts Management System to assist the sales team with data capturing and information management.

The sales team currently have an established and strict manual business process and workflow that they follow whenever they secure a new customer. A sales consultant collects information from a new customer and sends it to an Admin Assistant for data capturing. The Admin Assistant captures the customer information, opens the customer account and compiles a new customer report. Once finalized the new customer report is forwarded to a Sales Manager. In the sales management office, the Sales Manager receives the new customer report, confirms the new customer details, approves the new customer account opened by the Admin Assistant and ends the process by notifying the Sales Executive.

As part of the business rules, a customer will be allowed to open more than one account. Using the account, the customer would be allowed to list many products. The account is allowed to idle with no product listing. Products belong to certain categories or are categorised. All Sales consultants, Admin assistants and Sales managers are all permanent employees of Reciprocal Sales Corporation, and their information will be stored in one table on the database. The Sales consultant would be assigned to many customer accounts. Lastly, the account is allowed to idle with no Sales consultant assigned to it.

In terms of technology, the system will consist of a relational database based on an Oracle relational database management system (RDBMS) and will be developed using Java and other open-source platforms. The Software Development department at Reciprocal Sales Corporation has adopted agile project management instead of relying on the traditional approach to project management. The developers will handle their own Unit testing, and further testing on the system

will be outsourced to one of the leading Testing Consulting Company in the world because of the standards imposed by the Fortune 500 corporations that have entrusted Reciprocal Sales Corporation with their product business.

Q.4.1	Based on the Reciprocal Sales Corporation's current established and strict manual business processes and workflow outlined in the case study, develop a comprehensive UML activity diagram to model the workflows.	(20)
Q.4.2	Identify a minimum of three classes from the case study and draw a domain class diagram using the correct UML notation to present the associations amongst the classes. You do not have to include attributes for any of the classes. Please use UML notation	(6)
Q.4.3	The case study has numerous use cases, and detailed information about each use case is described with a use case description. List any three aspects of a use case covered in a use case description.	(3)
Q.4.4	In today's interconnected world, systems need reliable access control systems to keep the data secure. List and define the three elements that access control systems rely on.	(6)

[TOTAL MARKS: 100]